SILAS M. FALDE

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EDUCATION

University of Michigan School of Engineering — **Ann Arbor, MI**
Senior B.E.S. in Data Science, GPA: 3.09/4.00 — *August 2023 – December 2026*

Relevant Courses: Machine Learning, Artificial Intelligence, Computer Vision, Computational Modelling of Complex Systems, Database Management, Regression Analysis, Probability and Statistics, Bayesian Data Analysis, Data Structures and Algorithms

EXPERIENCE

Ally Financial, Inc. — **Detroit, MI**
Intern, Artificial Intelligence & Advanced Analytics — *May 2026 – August 2026*

- **Prototyped large language model workflows** to automate internal tasks and improve efficiency and safety.
- **Built Python NLP pipelines** to evaluate PII masking and redaction model performance for sensitive company documents.
- **Enhanced** a data dictionary migration tool by adding **batch processing**, smarter metadata selection, and a streamlined interface to improve usability and efficiency.

University of Michigan Department of Epidemiology — **Ann Arbor, MI**
Research Assistant – Computational Lead — *September 2025 – May 2026*

- Led development of a Python pipeline to classify collective-violence narratives using prompt-based large language model inference and structured response processing.
- Built reusable preprocessing utilities for NVDRS-style text, demographic derivation, path management, and annotation-round data handling.
- Designed human annotation workflows and measured inter-annotator agreement with Cohen's kappa and Krippendorff's alpha to resolve disagreements and assess model performance.

Ally Financial, Inc. — **Detroit, MI**
Intern, Data Management — *May 2025 – August 2025*

- **Created** data dictionaries and lineage documentation in **Python** to improve data governance and compliance.
- **Added data quality checks** to existing data pipelines with **Python and SQL** to improve data integrity and reliability.
- **Developed** a **Power BI** data availability dashboard to monitor refresh rates and availability for critical data assets.

University of Michigan Department of Epidemiology — **Ann Arbor, MI**
Research Assistant — *September 2024 – May 2025*

- Earned 3rd place in the CDC Youth Mental Health: Novel Variables Data Challenge, placing in roughly the top 1% of entries.
- Developed Python pipelines on a high-performance computing cluster to clean, normalize, and engineer features from more than 400,000 suicide-victim narratives.
- Used large language models to extract structured trends from narrative text and evaluated model-assisted outputs with classification metrics and error analysis.
- Documented findings with reproducible analyses, reports, and visualizations for the research team and publication.

PROJECTS

Home Server Infrastructure — **Ann Arbor, MI**
Systems Developer — *2024 – Present*

- **Designed and operate** a single-host **Ubuntu self-hosting platform** using **Docker Compose** stacks for ingress, media, cloud storage, AI services, monitoring, automation, and game hosting.
- **Automated media workflows** with **Bash and Python** for queueing, GPU-accelerated HEVC encoding, staging, metadata-safe publishing, library audits, and failure-aware batch processing.
- **Configured** Cloudflare Tunnel and Caddy ingress, Tailscale private administration, systemd scheduling, service health checks, and documented upgrade, backup, recovery, and security procedures.

FPS PvP Dynamics — **Ann Arbor, MI**
Project Developer — *March 2026 – April 2026*

- **Built** a configurable **agent-based simulation** in Python to model player movement, combat, objectives, respawning, and stochastic per-tick match dynamics.
- **Developed trace capture, metrics export, visualization, and automated tests** for debugging and validating simulation behavior.
- **Ran parallel parameter sweeps and replication studies**, **Latin hypercube sampling**, and **Sobol sensitivity analysis** to quantify the effect of model parameters.

Photo Processing Helper — **Ann Arbor, MI**
Project Developer — *Jan 2026 - Present*

- **Built a modular Python CLI and library** for photo framing, collage generation, panorama stitching, and Nikon RAW-to-JPG matching across large image folders.
- **Implemented OpenCV, NumPy, Pillow, and rawpy pipelines** for resizing, cropping, alignment, blending, and exact-dimension output while preserving high-bit-depth detail.
- **Added tests and validation** for file discovery, output integrity, stitching behavior, and repeatable batch processing.

Predicting Survival in the ICU — **Ann Arbor, MI**
Project Developer — *January 2025 – March 2025*

- **Developed** logistic-regression, convolutional, and recurrent-neural-network models in **Scikit-learn and PyTorch** to predict ICU mortality from clinical time-series and static features.
- **Built reproducible data loaders and preprocessing** for missing values, temporal observations, and engineered clinical features, with configuration-driven training and evaluation.
- **Evaluated models** with **AUROC**, visualization, and automated tests covering CNN, transformer, and output behavior.

SKILLS

Programming: Python, C++, Bash, JavaScript, R, SQL
Machine Learning: PyTorch, Scikit-learn, NumPy, Pandas, Matplotlib, OpenCV, spaCy, NLTK
Web: HTML, CSS, JavaScript, responsive design, static-site publishing
Systems: Linux, Docker Compose, systemd, CMake, Make, Git, CI/CD
Software: VS Code, Microsoft Office Suite, Google Workspace, Snowflake, Collibra, Alation, Manta